

Soluzione Esercizi 1.5.2 algebra di Boole

1 Funzione a 2 variabili

A	B	F	G	H	I
0	0	1	1	1	1
0	1	0	1	0	1
1	0	1	1	1	0
1	1	1	1	0	0

Svolgimento delle Espressioni Logiche

$$F(A, B) = A \cdot \overline{AB} + \bar{B} \iff A(A + \bar{B}) + \bar{B} \iff AA + A\bar{B} + \bar{B} \iff A + A\bar{B} + \bar{B} \iff A(1 + \bar{B}) + \bar{B} \iff \boxed{A + \bar{B}}$$

$$\begin{aligned} G(A, B) &= \bar{A} + B + A\bar{B} + \overline{A + \bar{B}} \iff \bar{A} + B + A\bar{B} + \bar{A}B \\ &\iff \bar{A}(1 + B) + B + A\bar{B} \iff \bar{A} + \underbrace{B + A\bar{B}}_{\text{Distributiva}} \iff \bar{A} + (B + A) \cdot (B + \bar{B}) \iff \bar{A} + B + A \iff 1 + B \iff \boxed{1} \end{aligned}$$

$$H(A, B) = \overline{AB\bar{B}} \iff \overline{(\bar{A} + B)\bar{B}} \iff \overline{AB + B} \iff \overline{B(1 + \bar{A})} \iff \boxed{\bar{B}}$$

$$I(A, B) = \bar{A} + \overline{A + \bar{A}\bar{B}} \iff \bar{A} + \overline{A + \bar{A} + \bar{B}} \iff \bar{A} + \overline{(1 + \bar{B})} \iff \bar{A} + \bar{1} \iff \bar{A} + 0 \iff \boxed{\bar{A}}$$

2 Funzioni a 3 variabili

A	B	C	F	G	H	I	L	M
0	0	0	0	0	0	1	0	0
0	0	1	0	0	0	1	0	1
0	1	0	1	0	0	1	0	1
0	1	1	1	0	0	0	0	1
1	0	0	1	1	1	1	0	1
1	0	1	1	0	1	1	1	1
1	1	0	1	0	1	1	1	1
1	1	1	1	0	1	1	1	1

Svolgimento delle Espressioni Logiche

$$F(A, B, C) = CB + A + B \iff B(C + 1) + A \iff B + A \iff \boxed{A + B}$$

$$G(A, B, C) = A \cdot A\bar{B} \cdot \overline{ABC} \iff A\bar{B} \cdot (\bar{A} + B + \bar{C}) \iff \underbrace{A\bar{B}\bar{A}}_0 + \underbrace{A\bar{B}B}_0 + A\bar{B}\bar{C} \iff \boxed{A\bar{B}\bar{C}}$$

$$H(A, B, C) = \bar{A} + \overline{A + CBA} \iff \bar{A} + \bar{A} \cdot \overline{CBA} \iff \bar{A} + \underbrace{\bar{A} \cdot \bar{C}\bar{B} \cdot \bar{A}}_0 \iff \bar{A} \iff \boxed{\bar{A}}$$

$$\begin{aligned} I(A, B, C) &= AC + A\bar{B} + \overline{C(A + B)} \iff AC + A\bar{B} + \overline{CA + CB} \\ &\iff AC + A\bar{B} + (\bar{C} + \bar{A})(\bar{C} + \bar{B}) \iff AC + A\bar{B} + \underbrace{\bar{C}\bar{C}}_{\bar{C}} + \bar{C}\bar{B} + \bar{A}\bar{C} + \bar{A}\bar{B} \end{aligned}$$

$$\iff AC + \underline{A\bar{B}} + \underline{\bar{C}} + \bar{C}\bar{B} + \bar{A}\bar{C} + \bar{A}\bar{B} \iff \bar{B} \underbrace{(A + \bar{A})}_1 + \bar{C} \underbrace{(1 + \bar{B})}_1 + AC + \bar{A}\bar{C}$$

$$\iff \bar{B} + \underline{\bar{C}} + AC + \underline{\bar{A}\bar{C}} \iff \bar{C} \underbrace{(1 + \bar{A})}_1 + \bar{B} + AC \iff \bar{C} + \bar{B} + AC$$

$$\iff \bar{B} + (\bar{C} + A) \underbrace{(\bar{C} + C)}_1 \iff \boxed{\bar{B} + \bar{C} + A}$$

$$L(A, B, C) = (A + \bar{B})(A + B)(\bar{A} + B + C) \iff (A + \underbrace{B \cdot \bar{B}}_0)(\bar{A} + B + C)$$

$$\iff A\bar{A} + AB + AC \iff BA + AC \iff \boxed{A(B + C)}$$

$$\begin{aligned}
M(A, B, C) &= (A + B + C)(\bar{A} + \bar{B} + \bar{C}) + B \\
&\iff \underbrace{A\bar{A}}_0 + A\bar{B} + A\bar{C} + B\bar{A} + \underbrace{B\bar{B}}_0 + B\bar{C} + C\bar{A} + C\bar{B} + \underbrace{C\bar{C}}_0 + B \\
&\iff A\bar{B} + A\bar{C} + B\bar{A} + \underbrace{B\bar{C}}_0 + C\bar{A} + C\bar{B} + B \\
&\iff B(1 + \bar{C}) + A\bar{B} + A\bar{C} + B\bar{A} + C\bar{A} + C\bar{B} \iff \underline{B} + A\bar{B} + A\bar{C} + \underline{B\bar{A}} + C\bar{A} + C\bar{B} \\
&\iff B + A\bar{B} + A\bar{C} + C\bar{A} + C\bar{B} \iff \underbrace{(B + \bar{B})}_1 (B + A) + A\bar{C} + C\bar{A} + C\bar{B} \\
&\iff A + B + A\bar{C} + C\bar{A} + C\bar{B} \iff A + B + A\bar{C} + C\bar{A} \\
&\iff \underbrace{(A + \bar{A})}_1 (A + C) + \underbrace{(B + \bar{B})}_1 (B + C) \iff A + C + B + \bar{C} \iff \boxed{A + B + C}
\end{aligned}$$

3 Funzioni a 4 variabili

A	B	C	D	F	G	I	AB	BC	AC	(AB+BC+AC)	H
0	0	0	0	0	1	0	0	0	0	0	0
0	0	0	1	0	1	0	0	0	0	0	0
0	0	1	0	1	1	1	0	0	0	0	0
0	0	1	1	1	1	1	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0	0
0	1	0	1	0	1	0	0	0	0	0	0
0	1	1	0	1	1	1	0	1	0	1	0
0	1	1	1	1	0	1	0	1	0	1	1
1	0	0	0	1	1	1	0	0	0	0	0
1	0	0	1	1	1	1	0	0	0	0	0
1	0	1	0	1	1	1	0	0	1	1	0
1	0	1	1	1	1	1	0	0	1	1	1
1	1	0	0	0	1	1	1	0	0	1	0
1	1	0	1	0	1	1	1	0	0	1	1
1	1	1	0	1	1	1	1	1	1	1	0
1	1	1	1	1	1	1	1	1	1	1	1

Svolgimento delle Espressioni Logiche

$$\begin{aligned}
F(A, B, C, D) &= CD + A\bar{B} + \overline{A\bar{B}} + \bar{A} + C\bar{D} \iff \\
&\iff CD + A\bar{B} + A(\bar{B}\bar{A}) + C\bar{D} \iff \boxed{CD + A\bar{B} + A\bar{B} + C\bar{D}} \\
&\iff C(D + \bar{D}) + A\bar{B} + A\bar{B} \iff C + A\bar{B} + A\bar{B} \iff \boxed{C + A\bar{B}} \\
\\
G(A, B, C, D) &= \bar{B} + AB + \overline{CD} + \bar{C} \iff (\bar{B} + A) \underbrace{(\bar{B} + B)}_1 + (\bar{C} + C) \underbrace{(\bar{D} + C)}_1 \iff \\
&\iff \boxed{\bar{B} + A + \bar{D} + \bar{C}} \\
\\
H(A, B, C, D) &= \bar{A}BCD + AB\bar{C}D + A\bar{B}CD + ABCD \iff \\
&\iff BCD \underbrace{(A + \bar{A})}_1 + A\bar{B}CD + AB\bar{C}D \iff BCD + A\bar{B}CD + AB\bar{C}D \iff \\
&\iff BD \underbrace{(C + A\bar{C})}_{(C+\bar{C})(A+C)} + A\bar{B}CD \iff BD(A + C) + A\bar{B}CD \iff BDA + BDC + A\bar{B}CD \iff \\
&\iff D(BA + \underbrace{BC + A\bar{B}C}_{C(B+A\bar{B})}) \iff D(BA + C \underbrace{(B + A\bar{B})}_{(B+A)(\bar{B}+B)}) \iff \\
&\iff D(BA + C(B + A)) \iff D(BA + CB + CA) \iff \\
&\iff \boxed{D(B(A + C) + AC)} \vee \boxed{D(BA + BC + AC)} \iff I(A, B, C, D) = \boxed{A + C}
\end{aligned}$$